Module -4 :: Capital Budgeting

Introduction to capital

Capital forms the base for the business. Capital, in general, does not mean only money. It may refer to money's worth also. Capital has different forms. Creativity, innovation or new ideas can be considered as one form of capital. Some people have ideas but they may not have money. There are some others who have money only. The ideal combination for business is to have both. Today, there are different sources of raising finance for many types of business provided we have the margin or the base money.
It is the capital that keeps any business going on. There are a number of instances where the business is closed for want of capital. In most of these cases, the probable reasons could be mismanagement of capital or lack of information about the different sources of capital or factors to be considered to estimate the capital requirements.

Meaning of capital

Capital is defined as wealth, which is created over a period of time through abstinence to spend. There are different forms of capital: property, cash or titles to wealth. It is the aggregate of funds used in the short-run and long-run. An economist views capital as the value of total assets available with the business.
An accountant sees the capital as the difference between the assets and liabilities. In this chapter, we consider capital as the total amount of finances required by the business to conduct its business operations both in the short-run and long-run.

Nature of capital

In an organization, **capital** refers to the financial resources and assets that are used to fund operations, investments, and growth. It is a critical factor for the sustainability and expansion of a business. Here are the key features of capital in an organization:

1. Source of Funding for Operations

- **Feature:** Capital provides the financial resources required to run the daily operations of a business.
- **Example:** A restaurant uses capital to purchase ingredients, pay employees, and cover utilities, allowing it to operate smoothly and serve customers.

2. Facilitates Growth and Expansion

- **Feature:** Capital is essential for enabling an organization to grow and expand its business operations.
- **Example:** A software company uses venture capital to hire more developers, invest in marketing, and launch its product in international markets.

3. Enables Investment in Assets

- **Feature:** Organizations use capital to acquire long-term assets such as machinery, equipment, or property that support business functions.
- **Example:** A manufacturing company uses capital to purchase new machinery to improve production efficiency and increase output.

4. Provides Risk Management

- **Feature:** Sufficient capital acts as a cushion to absorb losses or unexpected expenses, helping organizations manage risks.

- **Example:** During a downturn, a retail store relies on its capital reserves to continue paying employees and covering overheads while waiting for better sales conditions.

5. Supports Innovation and R&D

- **Feature:** Capital is required for research and development (R&D) to create new products or improve existing ones.
- **Example:** Apple invests capital into R&D to develop new features for its iPhones and other products, maintaining its competitive edge in the technology market.

6. Enhances Competitive Advantage

- **Feature:** Capital enables businesses to invest in marketing, technology, and other initiatives that can give them a competitive advantage.
- **Example:** Nike uses capital to sponsor athletes and run global advertising campaigns, strengthening its brand and attracting more customers worldwide.

7. Allows for Debt Financing

- **Feature:** Capital is necessary to access additional funding through loans or bonds, enabling businesses to undertake large projects or investments.
- **Example:** A real estate developer uses its existing capital to secure a loan for a new building project, leveraging debt to finance the construction.

8. Drives Strategic Investments

- **Feature:** Capital allows organizations to make strategic investments that support their long-term goals and objectives.
- **Example:** Google invests capital in acquiring other companies (such as YouTube) to diversify its portfolio and enhance its market position in different sectors.

9. Enables Profit Generation

- **Feature:** The deployment of capital in various business activities, such as production or marketing, drives revenue generation and profit growth.
- **Example:** A car manufacturer invests in advertising campaigns and production improvements, increasing vehicle sales and, subsequently, company profits.

10. Supports Financial Stability

- **Feature:** Adequate capital helps ensure an organization maintains financial stability, even during periods of low cash flow or unexpected challenges.
- **Example:** During an economic recession, a logistics company's strong capital position allows it to continue operations while competitors may struggle or go bankrupt.

These features of capital are integral to the overall success and sustainability of an organization. They influence everything from daily operations to long-term strategic decisions and financial health.

Significance of capital

Following are the key points that highlight the significance of capital in an organization:

1. Foundation for Operations

- Capital provides the essential funds needed for day-to-day business operations, including paying for raw materials, wages, utilities, and other essential expenses.
- **Example:** A small retail store requires capital to purchase inventory, pay for utilities, and cover employee wages. Without sufficient operating capital, the store cannot stay open or stock products.

2. Enables Growth and Expansion

- Adequate capital allows an organization to invest in new markets, expand its product offerings, and scale operations, fueling business growth and long-term success.
- **Example:** A tech startup uses capital from venture capitalists to expand its software development team and launch a new version of its app, increasing its customer base and entering new markets.

3. Financial Stability

- Capital acts as a financial cushion, providing a buffer against unexpected costs, economic downturns, or business disruptions, helping ensure the business remains solvent.
- **Example:** During an economic downturn, a construction company relies on its capital reserves to continue paying its employees and suppliers while waiting for new projects to be secured, ensuring it avoids layoffs or closure.

4. Supports Research and Innovation

- Capital is crucial for investing in research and development (R&D), enabling an organization to innovate, develop new products, and maintain competitiveness in the market.
- **Example:** Pharmaceutical companies like Pfizer use capital to fund R&D efforts for new drug development. This capital investment allows them to discover innovative treatments that could lead to high market demand and profitability.

5. Increases Profitability Potential

- By investing capital in assets, technology, and marketing, organizations can increase efficiency, reach more customers, and ultimately enhance profitability.
- **Example:** An e-commerce business invests capital in improving its website and customer service infrastructure, which results in increased customer satisfaction, higher conversion rates, and a boost in sales.

6. Risk Management and Contingencies

- Sufficient capital helps an organization manage risks and handle unexpected events, such as economic recessions or market fluctuations, without jeopardizing operations.
- **Example:** An airline company keeps a portion of its capital in reserve to manage potential risks such as a sudden spike in fuel prices or a global health crisis, like the COVID-19 pandemic, ensuring the business can survive difficult times.

7. Capital for Debt Financing

- Capital from investors or equity financing can be used to secure loans or issue bonds, which helps organizations fund large projects or acquisitions without depleting cash reserves.
- **Example:** A real estate developer uses equity capital to raise funds from banks and issue bonds to finance the construction of a new high-rise apartment building, reducing the financial burden of relying on just one funding source.

8. Attracts Investors and Stakeholders

- Strong capital structure and financial health increase an organization's appeal to potential investors, venture capitalists, and business partners, providing access to more funding opportunities.
- **Example:** Tesla's strong capital structure and consistent growth have made it an attractive investment opportunity for stakeholders and institutional investors, which helped the company secure additional funding for expansion and innovation.

9. Supports Competitive Advantage

- Capital enables organizations to invest in marketing, branding, technology, and human resources, all of which are essential for building and maintaining a competitive edge in the market.
- **Example:** Nike invests capital in advanced manufacturing technologies and marketing campaigns, strengthening its brand presence globally and staying ahead of competitors in the sports apparel market.

10. Improves Organizational Valuation

- The level of capital an organization has can significantly influence its market valuation, creditworthiness, and ability to secure favorable terms in business transactions.
- **Example:** A well-capitalized company like Amazon can invest in infrastructure, such as fulfillment centers and logistics networks. As a result, its valuation continues to rise because of its ability to scale quickly and meet customer demand efficiently.

Types of Capital

Capital can broadly be divided into two types: "fixed capital and working capital."

Fixed Capital

Fixed capital is that portion of capital which is invested in acquiring long-term assets such as land and buildings, plant and machinery, furniture and fixtures, and so on. Fixed capital forms the skeleton of the business. It provides the basic assets as per the business needs. These assets are not meant for resale. They are intended to generate revenues.

The following are the features of fixed assets:

1. Permanent in nature Fixed capital is more or less permanent in nature. It is generally not withdrawn as long as the business carries on its business.
2. Profit generation Fixed assets are the sources of profits but they can never generate profits by themselves. They use stocks, cash and debtors to generate profits.
3. Low liquidity The fixed assets cannot be converted into cash quickly. Liquidity refers to conversion of assets into cash.
4. Amount of fixed capital The amount of fixed capital of a company depends on a number of factors such as size of the company, nature of business, method of production and so on. A manufacturing company such as steel factory may require relatively large finance when compared to a service organisation such as a software company.
5. Utilised for promotion and expansion The fixed capital is mostly needed at the time of promoting the company to purchase the fixed assets or at the time of expansion/modernisation. In other words, the need for fixed capital arises less frequently.

Types of Fixed Assets

Fixed assets can be divided into three types:
1. Tangible Fixed Assets These are physical items which can be seen and touched. Most of the common fixed assets are land, buildings, machinery, motor vehicles, furniture and so on.
2. Intangible Fixed Assets These do not have physical form. They cannot be seen or touched. But these are very valuable to business. Examples are goodwill, brand names, trademarks, patents, copy rights and so on.

3. Financial Fixed Assets These are investments in shares, foreign currency deposits, government bonds, shares held by the business in other companies and so on.

Working Capital

Working capital is the flesh and blood of the business. It is that portion of capital that makes a company work. It is not just possible to carry on the business with only fixed assets; working capital is a must. Working capital is also called circulating capital. It is used to meet regular or recurring needs of the business. The regular needs refer to the purchase of materials, payment of wages and salaries, expenses like rent, advertising, power and so on. In short, working capital is the amounts needed to cover the cost of operating the business.

Working capital takes the form of cash, near cash and other assets in the process of moving towards the cash form in a short period. These other assets are stocks of raw materials, supplies needed for manufacture, stocks of finished goods ready for sale, semi-processed items or components, debtors and other short-term investments if any.

Types of working capital

Working capital can be categorized into different types based on the purpose, duration, and nature of the funds needed for business operations. These types help organizations manage their short-term financial needs effectively. Here are the main types of working capital:

1. Permanent (or Fixed) Working Capital

- **Definition:** This is the minimum amount of working capital that an organization needs to maintain its day-to-day operations throughout the year. It remains relatively constant over time.
- **Example:** A manufacturing company may always need a certain amount of capital to cover basic operations such as inventory and employee wages, regardless of seasonal fluctuations.
- **Purpose:** It covers the core operations of the business that do not fluctuate significantly, such as essential materials, production processes, and routine expenses.

2. Temporary (or Variable) Working Capital

- **Definition:** Temporary working capital is the additional capital required to meet seasonal fluctuations, unexpected demands, or business expansions.
- **Example:** A clothing retailer needs extra capital during the holiday season to purchase more inventory and handle increased customer demand, which is temporary and not required year-round.
- **Purpose:** It helps businesses adjust to changes in demand, production schedules, or market conditions that are not permanent.

3. Gross Working Capital

- **Definition:** Gross working capital refers to the total capital invested in the current assets of a business (i.e., cash, accounts receivable, inventory, etc.) that are used to fund day-to-day operations.
- **Example:** A wholesale distributor's gross working capital would include the value of all its inventories, outstanding receivables, and any cash reserves available for short-term operations.
- **Purpose:** It gives a picture of the total amount of capital tied up in assets that are required to run day-to-day operations.

4. Net Working Capital

- **Definition:** Net working capital is the difference between a company's current assets and current liabilities. It indicates the short-term liquidity available to fund operations and cover debts due in the next year.
- **Formula: Net Working Capital = Current Assets - Current Liabilities**
- **Example:** If a company has $500,000 in current assets (inventory, receivables, cash) and $300,000 in current liabilities (short-term debt), its net working capital would be $200,000.
- **Purpose:** Net working capital is a key indicator of a company's financial health and its ability to pay off short-term obligations.

5. Regular (or Routine) Working Capital

- **Definition:** This refers to the working capital required for regular, ongoing business operations that occur routinely, such as paying suppliers and covering salaries.
- **Example:** A bakery uses regular working capital to buy raw materials, pay workers, and manage other daily costs that are consistent each month.
- **Purpose:** It ensures the smooth functioning of daily operations without disruptions.

6. Special Working Capital

- **Definition:** This type of capital is required for specific or extraordinary events, such as business expansions, launching a new product, or handling an unforeseen surge in demand.
- **Example:** A software company may need special working capital to finance the launch of a new version of its software, including marketing and additional development costs.
- **Purpose:** It is used to fund specific projects or one-time events that temporarily affect the business.

7. Cyclic Working Capital

- **Definition:** Cyclic working capital is the capital required for businesses that experience cyclical variations in demand, production, and sales, such as in industries that are affected by seasons or economic cycles.
- **Example:** A toy company experiences increased sales during the holiday season, requiring additional working capital to buy inventory and hire temporary workers during that peak period.
- **Purpose:** It supports the business during cyclical peaks and helps cover the additional costs that arise during high-demand periods.

8. Negative Working Capital

- **Definition:** Negative working capital occurs when a company's current liabilities exceed its current assets. This indicates potential liquidity problems or that the business is operating on credit from suppliers.
- **Example:** A grocery chain may have negative working capital because it relies on supplier credit and quick inventory turnover, allowing it to run with minimal cash flow.
- **Purpose:** While negative working capital can signal liquidity issues, it may also indicate a very efficient business model, especially in industries where companies manage short-term debt well.

9. Reserve Working Capital

- **Definition:** This type refers to the additional working capital set aside as a buffer for unexpected financial challenges or business opportunities.
- **Example:** A tech startup may set aside reserve working capital to cover potential funding gaps if there are delays in customer payments or unforeseen operational costs.
- **Purpose:** It provides a financial cushion to prevent disruptions and ensures business continuity during tough times.

Each of these types of working capital serves a specific purpose depending on the organization's operations, growth stage, and industry. Effective working capital management ensures a business can meet its short-term obligations, manage growth, and avoid financial strain.

Features of Working Capital

1. Short life span Working capital changes in its form: from cash to stock; stock to debtors; debtors to cash. The cash balances may be kept idle for a week or so, debtors have a life span of a few months, raw materials are held for a short-time until they go into production; finished goods are held for a short-time until they are sold.
2. Smooth flow of operations Adequate amount of working capital enables the business to conduct its operations smoothly. It is, therefore, called the "flesh and blood" of the business.
3. Liquidity The assets represented by the working capital can be converted into cash quickly within a short period of time unlike fixed assets.
4. Amount of working capital The amount of working capital of a business depends on many factors such as size and nature of the business, production and marketing policies, business cycles and so on. These are discussed in detail in the following section.
5. Utilised for payment of current expenses The working capital is used to pay for current expenses such as suppliers of raw materials, payment of wages and salaries, rent and other expenses and so on.

Components of Working Capital

From the accounting point of view, working capital is the difference between current assets and current liabilities. (Working capital = Current Assets - Current Liabilities)

Current Assets

Cash is required to pay salaries, office expenses and to pay creditors for purchases.
Stock of raw materials in adequate quantities to ensure uninterrupted production.
Stock of finished goods in sufficient quantities to meet the demand from customers.
Debtors, that is, people to whom we sell goods on credit basis for increased sales.
Prepaid expenses, that is, the expenses paid in advance such as insurance, rent, salaries and so on.
Bills receivables these are the bills of exchange received for the money lent or to be received for a short period.

Current Liabilities

Creditors, that is, the people from whom we purchase on credit basis.
Accruals, that is, those expenses in respect of which, the liability has arisen. In other words, the expenses have fallen due and hence to be incurred, such as interest, salaries, taxes and so on.
Bills payables these are the bills of exchange against which money is to be paid within a short period.

Working capital cycle

Given below is a model of the working capital cycle (see Fig. 15.1). From the above working capital cycle, it can be seen that raw material is bought by paying cash or by giving a bill. (If the goods are purchased against a bills payable, cash will be paid against the bill at a later date as mentioned in the bill.)
Raw material is processed into finished goods and these are sold either for cash or for a bill. If the goods are sold against a bills receivable, cash will be received at a later date as mentioned in the bill.

The flow of money in the business is referred to as "working capital cycle" and is vital for smooth functioning of any business. The money invested in working capital is continually moving towards purchase of material inputs, wages and salaries and other expenses.

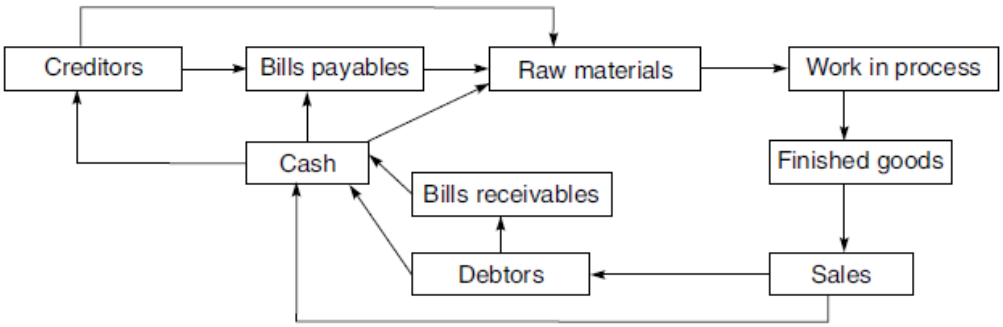

Fig. 15.1 A Working Capital Cycle

How Much to Invest in Working Capital?

This is a critical question. If more capital is earmarked, it may affect the profitability. Similarly, if less is set aside, it may affect the liquidity of the business. For an optimal approach, it is necessary to determine the minimum funds, study the pattern of receipts and payments, and plan in advance the short-term borrowings and investments.

Factors governing working capital requirements

The **requirements of working capital** in an organization depend on several factors that influence how much capital is needed for daily operations. Here are **10 key factors determining the requirements of working capital**, along with examples for each:

1. Nature of Business

- **Explanation:** The type of business and industry plays a significant role in determining working capital needs. Businesses that deal with perishable goods or require continuous production may need more working capital.
- **Example:** A grocery store dealing with fresh produce requires more working capital to maintain inventory compared to a consulting firm with minimal inventory.

2. Business Cycle

- **Explanation:** The length and nature of a company's business cycle, including production and sales cycles, impact working capital needs. A longer cycle means the business will need more capital to fund operations during the waiting period.
- **Example:** A construction company with long project timelines needs more working capital to cover labor, equipment, and material costs before receiving payment, compared to a fast-moving retail store.

3. Production Cycle

- **Explanation:** The time taken to convert raw materials into finished goods and then sell them affects the amount of working capital required. A longer production cycle means more working capital to support ongoing operations.
- **Example:** A car manufacturer needs substantial working capital to pay for parts and assembly labor while vehicles are being produced and before they are sold.

4. Seasonal Demands

- **Explanation:** Businesses with seasonal fluctuations in demand (such as retail or agriculture) need to adjust their working capital based on peak and off-peak seasons.

- **Example:** A toy store during the holiday season may need increased working capital to purchase inventory in bulk, while during the off-season, it needs less capital.

5. Credit Policy

- **Explanation:** The company's policy on offering credit to customers impacts its working capital needs. A generous credit policy leads to longer collection periods, requiring more working capital to finance outstanding receivables.
- **Example:** A furniture retailer that offers 90-day credit to customers will need more working capital compared to a business that only accepts cash or short-term payments.

6. Inventory Management

- **Explanation:** The amount of inventory an organization holds, along with the efficiency of inventory management, affects working capital. Businesses with high stock levels may need more working capital.
- **Example:** An electronics store that stocks a wide range of products needs substantial working capital to maintain inventory, while a small boutique with limited items needs less capital.

7. Sales Volume

- **Explanation:** Higher sales volumes generally require more working capital to handle the increased production, inventory, and receivables.
- **Example:** A mobile phone manufacturer experiencing a surge in sales during the launch of a new model needs additional working capital to meet the increased demand for raw materials, labor, and finished goods.

8. Profitability

- **Explanation:** More profitable companies may have more internal cash flow, reducing their dependence on external working capital financing. Less profitable companies may need more external capital to support operations.
- **Example:** A highly profitable software company can fund its operations with minimal working capital since it generates significant cash flow from its subscriptions. In contrast, a startup with low profitability may need additional working capital.

9. Supplier Credit

- **Explanation:** The amount of credit extended by suppliers impacts working capital requirements. Longer supplier credit terms can reduce the need for working capital, as payments to suppliers can be delayed.
- **Example:** A manufacturing company that has negotiated extended payment terms with suppliers (e.g., 60 days) will need less working capital than one that must pay suppliers immediately upon receiving goods.

10. Inflation and Price Fluctuations

- **Explanation:** Inflation and price fluctuations in raw materials or goods affect the cost of production and inventory, thereby influencing the working capital requirement. If prices rise, businesses may need more working capital to maintain inventory levels.
- **Example:** A construction company might require additional working capital during periods of inflation when the cost of building materials like steel and cement increases, requiring the business to allocate more funds for the same amount of materials.

These factors, in combination, help organizations determine the optimal amount of working capital needed to maintain smooth operations. Businesses must continuously assess these elements to ensure they have sufficient resources to meet their short-term financial obligations and support growth.

Sources of finance

The following are the common sources of finance:

- Long-term finance
- Medium-term finance
- Short-term finance

Now we will discuss each of these methods identifying the sources under each method:

I. LONG-TERM SOURCES

Long-term sources of finance are used by organizations to fund investments and projects that have a longer repayment period, typically exceeding 5 years. These sources help businesses finance capital-intensive projects like infrastructure, expansion, and long-term strategic growth. Below are the common **long-term sources of finance** for organizations, along with examples for each:

1. Equity Capital (Shares)

- **Definition:** Equity capital is the money raised by issuing shares to investors. Investors buy shares in exchange for partial ownership in the company, and they may receive dividends as a return on their investment.
- **Example:** A publicly listed company issues additional shares to raise funds for expanding its production capacity or entering a new market.
- **Purpose:** Equity capital provides a long-term source of funding without the obligation to repay or pay interest. However, it results in ownership dilution.

2. Long-Term Loans

- **Definition:** Long-term loans are borrowed funds from financial institutions (such as banks) that are paid back over a period exceeding 5 years, typically with a fixed or variable interest rate.
- **Example:** A real estate development company takes out a 10-year loan to finance the construction of a new shopping mall.
- **Purpose:** Long-term loans are used for capital-intensive projects and business expansions, where the repayment period is spread over several years.

3. Bonds (Corporate Bonds)

- **Definition:** Bonds are debt securities issued by an organization to raise long-term capital. Bonds have fixed or floating interest rates and are repaid at maturity, typically in 5 years or more.
- **Example:** A utility company issues 10-year bonds to fund the construction of a new power plant, with interest paid to bondholders annually.
- **Purpose:** Bonds are a common source of long-term financing, allowing businesses to raise large sums of money from investors with a commitment to repay over time.

4. Debentures

- **Definition:** Debentures are a type of long-term debt instrument issued by a company, usually secured by the company's assets. These offer fixed interest payments over a long duration.
- **Example:** A corporation issues 15-year debentures to raise funds for expanding its production facilities, offering an annual interest rate to investors.

- **Purpose:** Debentures provide companies with a large sum of capital for long-term use, with relatively fixed costs in terms of interest payments.

5. Venture Capital

- **Definition:** Venture capital is equity investment provided by venture capitalists (VCs) to startups and growing businesses with high growth potential. It usually involves an exit strategy, such as an IPO or acquisition, in 5 to 10 years.
- **Example:** A technology startup receives venture capital to develop new software, with the VCs expecting to exit after 5 years through an acquisition.
- **Purpose:** Venture capital funds high-risk, high-reward businesses in their early stages and allows the company to scale rapidly.

6. Private Equity

- **Definition:** Private equity involves investment by private equity firms in established businesses, typically in exchange for ownership or control. The investment horizon is usually long-term, often lasting 5 to 7 years or more.
- **Example:** A private equity firm invests in a manufacturing company, providing capital for its growth and expecting a return on investment through a future sale or public offering.
- **Purpose:** Private equity provides capital for expansion, restructuring, or management buyouts and is a long-term financing option for established businesses.

7. Retained Earnings

- **Definition:** Retained earnings are profits that a company retains after paying dividends, which are reinvested in the business for expansion, development, or paying off long-term debt.
- **Example:** A profitable retail company uses its retained earnings to open new stores in different cities.
- **Purpose:** Retained earnings provide a long-term, internal source of financing, helping companies fund their own growth without taking on additional debt.

8. Grants (Government or Institutional Grants)

- **Definition:** Grants are non-repayable funds or resources provided by the government or institutions for specific purposes, such as research, development, or infrastructure projects.
- **Example:** A clean energy company receives a government grant for developing renewable energy technologies, with no requirement for repayment.
- **Purpose:** Grants are often used for specific projects that align with public or institutional objectives and do not require repayment.

9. Convertible Bonds

- **Definition:** Convertible bonds are debt instruments that can be converted into equity (shares) at a later date, typically after a few years. These bonds provide long-term financing but offer the potential for conversion into ownership.
- **Example:** A biotechnology company issues convertible bonds that can be converted into company shares after 5 years, allowing investors to become equity holders.
- **Purpose:** Convertible bonds offer a hybrid option for both debt and equity financing, providing the business with funds while offering investors the opportunity to convert debt into equity if the company performs well.

10. Sale and Leaseback

- **Definition:** A sale and leaseback arrangement allows a company to sell its assets (such as property or machinery) to raise capital and simultaneously lease the asset back from the buyer, continuing to use the asset while freeing up cash for other investments.
- **Example:** A company sells its office building to a financial institution but enters into a lease agreement to continue operating from the same building, receiving funds to invest in expansion.
- **Purpose:** This method frees up long-term capital tied to assets without losing the use of the property or equipment.

11. Securitization

- **Definition:** Securitization involves converting a company's assets (such as loans, receivables, or mortgages) into tradable securities that are sold to investors.
- **Example:** A bank securitizes its mortgage portfolio by bundling mortgages into securities and selling them to investors to raise long-term capital.
- **Purpose:** Securitization allows businesses to access capital by converting illiquid assets into cash, while investors receive regular returns based on the income from those assets.

These **long-term sources of finance** enable businesses to make substantial investments in their growth and expansion. They come with different levels of risk, cost, and ownership implications, allowing organizations to choose the most appropriate method based on their specific needs and goals.

II. MEDIUM-TERM SOURCES

Medium-term sources of finance refer to the funds that organizations can raise to meet financial needs for a period typically ranging from 1 to 5 years. These sources are used for financing projects or investments that require more time to pay off than short-term financing but less than long-term financing. Here are some common **medium-term sources of finance** for organizations, along with examples:

1. Term Loans

- **Definition:** A term loan is a fixed amount of money borrowed from a financial institution (such as a bank) with a set repayment schedule and interest rate, typically for a period between 1 to 5 years.
- **Example:** A manufacturing company may take out a 3-year term loan to purchase new machinery and equipment for its production line.
- **Purpose:** Term loans are often used for purchasing fixed assets or funding expansions.

2. Leasing

- **Definition:** Leasing is a method where a company rents an asset (such as equipment, machinery, or vehicles) for a fixed period in exchange for periodic payments.
- **Example:** A construction company may lease heavy machinery like bulldozers for a 3-year period, avoiding the high upfront cost of purchasing them.
- **Purpose:** Leasing allows businesses to use expensive assets without committing significant capital upfront, and payments can be spread over time.

3. Debentures

- **Definition:** Debentures are medium-term debt instruments issued by companies to raise funds. They have a fixed interest rate and a specific maturity period, usually ranging from 1 to 5 years.
- **Example:** A company issues a debenture with a 4-year maturity to raise funds for working capital or expansion purposes.
- **Purpose:** Debentures are useful for raising capital for various purposes while spreading the repayment over several years.

4. Trade Credit

- **Definition:** Trade credit is a financing arrangement where a supplier allows the organization to purchase goods or services on credit, with payment due in a medium-term period (typically 30 to 180 days).
- **Example:** A retailer buys inventory from a wholesaler with a 90-day payment term, allowing it to sell the goods before having to pay the supplier.
- **Purpose:** Trade credit is an essential source of finance for businesses to manage cash flow while keeping stock levels adequate.

5. Bank Overdraft (Medium-Term Overdraft Facility)

- **Definition:** A bank overdraft allows a business to withdraw more money than it has in its current account up to an agreed limit. This is often used for medium-term financing needs when working capital is insufficient.
- **Example:** A retail store with an overdraft facility can withdraw extra funds to finance inventory purchases before the next peak sales season.
- **Purpose:** Overdrafts are generally used for short to medium-term needs, providing flexibility in managing cash flow gaps.

6. Hire Purchase

- **Definition:** Hire purchase is a type of financing agreement where an organization buys an asset and pays for it in installments over a period, typically 1 to 5 years. Ownership is transferred to the business after all installments are paid.
- **Example:** A logistics company might purchase delivery trucks via a hire purchase agreement, paying over 3 years while using the trucks in its operations.
- **Purpose:** Hire purchase is commonly used for purchasing vehicles, machinery, and equipment without an immediate full payment.

7. Medium-Term Bonds

- **Definition:** Medium-term bonds are debt securities issued by companies with a maturity period between 1 and 5 years. They are typically issued to institutional investors and have a fixed interest rate.
- **Example:** A utility company issues 5-year bonds to raise funds for infrastructure development, with interest payments made annually.
- **Purpose:** Bonds are an effective way for businesses to raise substantial capital for projects, with repayment over a few years.

8. Convertible Loans

- **Definition:** A convertible loan is a type of financing where the loan can be converted into equity (shares) of the company at a later date, typically within a few years.
- **Example:** A technology startup raises funds through convertible loans, where the investors have the option to convert the debt into equity after 3 years if the company performs well.
- **Purpose:** This form of financing helps startups raise capital without immediately diluting ownership but provides an option for investors to convert to equity if the company grows.

9. Government Grants and Subsidies (Medium-Term)

- **Definition:** Some government programs offer grants or subsidies for specific business activities, such as innovation, environmental sustainability, or expansion, with medium-term repayment or conditions.

- **Example:** A renewable energy company receives a government grant to develop a new solar energy technology, with the condition to report progress and meet milestones over the next 3 years.
- **Purpose:** Grants and subsidies are non-repayable funds, but sometimes medium-term conditions must be met, such as achieving specific targets within a certain period.

10. Private Equity or Venture Capital (Medium-Term Investment)

- **Definition:** Private equity or venture capital investors may provide funding in exchange for ownership equity, typically with an exit strategy planned within a few years.
- **Example:** A fast-growing tech startup receives a medium-term investment from a venture capital firm with the expectation that the firm will sell its stake within 4 years, either through an IPO or acquisition.
- **Purpose:** Private equity or venture capital can help businesses with high growth potential fund their operations, research, and expansion plans without taking on debt.

These **medium-term sources of finance** provide businesses with the flexibility to fund projects and operations that require more time to complete than short-term financing can support, without the long commitment of long-term loans or bonds.

III. SHORT-TERM SOURCES

Short-term sources of finance are typically used by organizations to meet immediate or urgent funding needs for day-to-day operations. These sources are generally for periods less than 1 year and are crucial for managing liquidity and covering working capital requirements. Here are the common **short-term sources of finance** for organizations, along with examples for each:

1. Bank Overdraft

- **Definition:** A bank overdraft allows a company to withdraw more money from its bank account than it has available, up to an agreed limit. The company only pays interest on the amount overdrawn.
- **Example:** A retail store uses a bank overdraft to cover a temporary cash shortfall, especially when customer payments are delayed.
- **Purpose:** It provides quick and flexible access to cash for covering immediate financial needs without having to apply for a loan.

2. Trade Credit

- **Definition:** Trade credit occurs when a supplier allows a business to purchase goods or services on credit, with payment due at a later date (typically within 30, 60, or 90 days).
- **Example:** A wholesaler allows a retailer to buy inventory now and pay for it in 60 days, which helps the retailer maintain cash flow while selling the goods.
- **Purpose:** Trade credit is a common short-term financing tool that helps businesses manage cash flow by deferring payments for goods purchased.

3. Short-Term Loans

- **Definition:** Short-term loans are loans from financial institutions with a repayment period of less than a year. They usually have a higher interest rate compared to long-term loans.
- **Example:** A manufacturing company takes a short-term loan to pay for raw materials in anticipation of receiving payments from customers soon.
- **Purpose:** Short-term loans are used to bridge temporary cash flow gaps or finance urgent needs, such as inventory purchases or emergency repairs.

4. Factoring

- **Definition:** Factoring involves selling a company's accounts receivable (outstanding invoices) to a third party (factor) at a discount in exchange for immediate cash.
- **Example:** A textile company sells its unpaid invoices to a factoring company, receiving immediate cash to continue operations while the factor collects the payments.
- **Purpose:** Factoring helps businesses access cash quickly, especially if they have slow-paying customers or need liquidity to cover expenses.

5. Commercial Paper

- **Definition:** Commercial paper is a short-term, unsecured promissory note issued by large corporations to raise funds for short-term liabilities, typically with maturities of 30 to 270 days.
- **Example:** A large corporation issues commercial paper to meet its payroll or finance inventory during a seasonal peak.
- **Purpose:** Commercial paper is typically used by well-established companies to raise funds quickly, often at lower interest rates than bank loans.

6. Trade Bills

- **Definition:** Trade bills (or bills of exchange) are short-term financial instruments used to settle payment between businesses, where one party agrees to pay a certain amount at a later date.
- **Example:** A supplier issues a trade bill to a buyer, who agrees to pay the amount within 90 days after receiving the goods.
- **Purpose:** Trade bills allow businesses to defer payments and manage their working capital needs efficiently.

7. Cash Credit

- **Definition:** Cash credit is a short-term financing option provided by banks that allows businesses to borrow funds up to a pre-approved limit. Interest is charged only on the amount utilized.
- **Example:** A manufacturing company uses cash credit to pay for raw material purchases, with repayment expected within 3-6 months.
- **Purpose:** Cash credit offers a flexible solution for managing daily working capital requirements and covering operational expenses.

8. Customer Advances

- **Definition:** Customer advances involve receiving payment from customers in advance for goods or services that will be delivered in the future.
- **Example:** A software company receives advance payment from customers for licenses that will be delivered over the next 6 months.
- **Purpose:** Customer advances help businesses manage cash flow, especially if they need to finance the production of goods or services before delivery.

9. Inventory Financing

- **Definition:** Inventory financing involves using a company's inventory as collateral to obtain short-term loans or lines of credit.
- **Example:** A wholesale distributor uses its stock of unsold goods as collateral to secure a short-term loan to cover operational expenses.
- **Purpose:** This type of financing helps businesses maintain liquidity by turning inventory into cash without having to sell the goods immediately.

10. Payables Financing (Supply Chain Financing)

- **Definition:** Payables financing is a type of financing where a third-party financier helps a company pay its suppliers earlier than the due date, usually at a discounted rate, to improve cash flow.
- **Example:** A retail company partners with a financial institution to pay its suppliers early, taking advantage of early payment discounts and improving its supplier relationships.
- **Purpose:** This form of financing helps businesses manage supplier payments and take advantage of discounts, while suppliers benefit from faster payments.

11. Credit Cards

- **Definition:** Credit cards provide businesses with a revolving line of credit that can be used to pay for expenses, with the option to repay the balance over a short-term period (typically monthly).
- **Example:** A small business owner uses a business credit card to cover office supplies and other operational costs, paying off the balance within the grace period to avoid interest charges.
- **Purpose:** Credit cards offer a convenient and flexible way to finance small, short-term business expenses, especially for businesses with limited access to other financing options.

These **short-term sources of finance** are essential for businesses to manage daily operations, cover unexpected expenses, and ensure smooth cash flow management. They are typically easier to access compared to long-term financing but may come with higher interest rates or the need for quick repayment.

Introduction – Capital Budgeting

Every organisation, irrespective of its nature (profit-making or otherwise) or size (big or small), in the course of its functioning, usually acquires, upgrades, replaces the assets such as land and buildings, plant and machinery and so on. For each of these, there exist two or more alternatives, which need to be carefully evaluated on the basis of their costs and revenues. To improve the quality of our decisions, an understanding of the principles and practices of capital budgeting is essential.

Features

Capital budgeting refers to the process of planning and evaluating the long-term investments and projects of an organization, such as acquiring new equipment, expanding facilities, or launching a new product line. It involves assessing the potential profitability and financial impact of these investments to ensure they align with the company's strategic goals.

Here are **10 key features of capital budgeting**, along with examples for each:

1. Long-Term Focus

- **Feature:** Capital budgeting involves decisions related to long-term investments that will impact the company over an extended period, usually several years.
- **Example:** A car manufacturing company decides to invest in a new production line for electric vehicles, which will operate for 10 years.
- **Purpose:** The decision-making process considers the long-term benefits and risks associated with these projects.

2. Investment Evaluation

- **Feature:** Capital budgeting involves evaluating investment opportunities to determine whether they are worth pursuing based on their potential return on investment (ROI).
- **Example:** A retail company evaluates whether to invest in a new warehouse, comparing projected costs, revenues, and payback periods.
- **Purpose:** Ensures that only the most profitable and viable projects are undertaken.

3. Cash Flow Estimation

- **Feature:** Capital budgeting requires estimating future cash flows, including both inflows (revenues) and outflows (costs), to assess the financial feasibility of the investment.
- **Example:** A real estate development company projects rental income from a new office building and estimates construction and maintenance costs to calculate net cash flows over the project's life.
- **Purpose:** Helps determine the profitability of an investment by forecasting the cash flows that will be generated.

4. Risk and Uncertainty Consideration

- **Feature:** Capital budgeting accounts for the risks and uncertainties associated with long-term investments, considering potential changes in market conditions, regulations, and other external factors.
- **Example:** A pharmaceutical company invests in research and development for a new drug, factoring in the risk of regulatory approval or market acceptance.
- **Purpose:** Ensures that decision-makers understand the level of risk involved and are prepared for uncertainties.

5. Discounted Cash Flow (DCF) Analysis

- **Feature:** One of the primary methods used in capital budgeting is **discounted cash flow** (DCF) analysis, which adjusts future cash flows to their present value using a discount rate (usually the company's cost of capital).
- **Example:** A technology firm uses DCF analysis to determine the present value of future revenue from a new software product, accounting for time value of money.
- **Purpose:** Allows businesses to compare the value of money now with the value of money in the future, helping to assess whether a project is financially viable.

6. Payback Period

- **Feature:** The **payback period** is the time required for the initial investment in a project to be recovered through its cash inflows.
- **Example:** A manufacturing company evaluates the payback period for a new machine. It expects to recover the $500,000 investment in 3 years based on projected annual savings.
- **Purpose:** Provides an estimate of how quickly the business will recoup its investment, which can be important for businesses seeking faster returns.

7. Profitability Index

- **Feature:** The **profitability index** (PI) is the ratio of the present value of future cash flows to the initial investment. It is used to evaluate the relative profitability of different investment options.
- **Example:** A company evaluating multiple projects finds that Project A has a profitability index of 1.5 (for every $1 invested, $1.50 in present value is generated), making it more attractive than Project B, which has a PI of 1.2.
- **Purpose:** Helps prioritize projects by comparing the present value generated per dollar invested.

8. Cost of Capital

- **Feature:** The **cost of capital** represents the return that a company must earn on its investments to satisfy its investors or creditors. It is used as the discount rate in capital budgeting calculations.
- **Example:** A company plans to finance a new expansion with a mix of debt and equity. The cost of capital is determined to be 8%, and this rate is used in DCF analysis to assess the feasibility of the project.

- **Purpose:** Ensures that the company’s investments generate returns that are higher than the cost of obtaining funds.

9. Strategic Alignment

- **Feature:** Capital budgeting decisions must align with the company's long-term strategic objectives, such as market expansion, product diversification, or sustainability goals.
- **Example:** A beverage company invests in a new plant in an emerging market to align with its strategy of global expansion and tapping into new consumer bases.
- **Purpose:** Ensures that investments contribute to the company's broader goals, rather than just providing short-term financial returns.

10. Post-Implementation Review

- **Feature:** After a project is implemented, capital budgeting includes a post-implementation review to evaluate the actual performance of the investment against the original projections.
- **Example:** A construction firm evaluates the return on investment from a newly built office building by comparing actual rental income with initial forecasts.
- **Purpose:** Helps organizations learn from past investments, refine future capital budgeting processes, and ensure that decisions align with financial goals.

These **features of capital budgeting** provide a structured approach for organizations to make informed decisions on large-scale investments. Capital budgeting helps minimize risks, ensures optimal allocation of resources, and maximizes the long-term profitability of the company.

Significance of Capital Budgeting

Capital budgeting is a crucial process for businesses because it helps them evaluate and prioritize long-term investment projects. Here's a brief overview of its significance:

1. Informed Decision-Making

- Capital budgeting helps businesses make informed decisions about which projects to pursue, based on their expected returns and alignment with the company's strategic objectives.

2. Optimal Resource Allocation

- It ensures that a company allocates its limited financial resources effectively, prioritizing investments that will provide the highest return.

3. Long-Term Growth and Sustainability

- By focusing on long-term projects, capital budgeting supports the growth and sustainability of the organization by funding investments that will generate future profits and maintain competitive advantage.

4. Risk Management

- The process involves analyzing the risks and uncertainties associated with each investment, allowing businesses to assess the potential downside and make adjustments accordingly.

5. Cost Control

- Capital budgeting helps in controlling costs by evaluating the financial feasibility of a project before committing significant resources, reducing the likelihood of overspending.

6. Improved Cash Flow Planning

- It aids in predicting future cash flows from investments, allowing companies to plan for financing needs and manage working capital effectively.

7. Maximizing Shareholder Value

- Through careful selection of profitable investments, capital budgeting helps in maximizing the long-term value for shareholders by choosing projects that increase the overall wealth of the business.

8. Strategic Alignment

- Capital budgeting ensures that investment decisions align with the company's overall strategic goals, such as market expansion, diversification, or technological advancement.

9. Performance Evaluation

- Post-investment reviews allow organizations to assess the actual performance of the project against initial expectations, providing valuable insights for future decision-making.

10. Investor Confidence

- A well-executed capital budgeting process demonstrates to investors and stakeholders that the company is managing its resources wisely and pursuing financially sound growth opportunities.

In short, capital budgeting plays a critical role in helping organizations make strategic, financially sound decisions that drive long-term success. It ensures that investments align with business goals, maximize returns, and minimize risks.

Capital Budgeting Proposal

A business needs a new machine and has to make a choice between machine Y and machine Z. The initial cost and the net cash flow over five years (income less running expenses but not depreciation) to the business have been calculated for each machine as follows:

	Machine Y (Rs)	Machine Z (Rs)
Initial cost	20,000	28,000
Net cash flow		
1	8,000	10,000
2	12,000	12,000
3	9,000	12,000
4	7,000	9,000
5	6,000	9,000

Only one machine is needed, and at the end of five years, the machine will have no value and will be scrapped. To finance the project the business can borrow money at 10 percent per annum. Which machine should be chosen?

This is a real situation confronting business with two alternatives. Sometimes the alternatives could be more. Which one is to be preferred?

These decisions are to be analysed based on their profitability. To understand this, let us see the different methods of investment appraisal.

Methods of Capital Budgeting

Capital budgeting decisions are made under different criteria. How are these criteria determined. These criteria differ in concepts. Some use thumb rules and some use logic and scientific approach. So, based on these criteria, the methods of capital budgeting can be classified as,

(a) Traditional methods
(i) Payback period
(ii) Accounting rate of return method

(b) Discounted cash flow methods.
(i) Internal rate of return (IRR) method
(ii) Net present value (NPV) method.
(iii) Profitability Index(PI) method.

Let us discuss these methods in detail.

1. Net Present Value (NPV)

Definition: NPV is the most widely used capital budgeting technique. It calculates the difference between the present value of cash inflows and the present value of cash outflows over the project's life.

Formula:

$$NPV = \sum \left(\frac{\text{Cash Inflow}_t}{(1+r)^t} \right) - \text{Initial Investment}$$

Where:

- t is the time period (usually in years).
- r is the discount rate (usually the company's cost of capital).
- Cash Inflow t is the expected cash inflow in year t.

Decision Rule:

- If NPV > 0, accept the project (it's expected to add value).
- If NPV < 0, reject the project (it's expected to destroy value).
- If NPV = 0, the project will break even.

- **Strengths**: Considers the time value of money and gives a direct indication of the project's value

2. Internal Rate of Return (IRR)

Definition: IRR is the discount rate that makes the NPV of the project equal to zero. It represents the expected rate of return on the investment.

Formula: The same as NPV, but IRR is the rate r that satisfies the equation:

$$\sum \left(\frac{\text{Cash Inflow}_t}{(1+\text{IRR})^t} \right) = \text{Initial Investment}$$

- **Decision Rule**:
  - If IRR > required rate of return (hurdle rate), accept the project.
  - If IRR < required rate of return, reject the project.
- **Strengths**: It gives an intuitive percentage that shows the project's profitability.

3. Payback Period

- **Definition**: The payback period measures the time it takes for a project to recover its initial investment from its cash inflows.
- **Formula**: The total initial investment is divided by annual cash inflows (if they are consistent over time).
- **Decision Rule**:
  - A shorter payback period is preferred because the investment is recovered faster, reducing the risk of the project.
  - It's often used as a rough measure of liquidity or risk.
- **Strengths**: Simple to calculate and understand.
- **Limitations**: Ignores the time value of money and cash inflows after the payback period.

4. Profitability Index (PI)

- **Definition**: The profitability index is the ratio of the present value of future cash inflows to the initial investment. It indicates how much value is created for each dollar invested.
- **Formula:**

$$PI = \frac{\sum \left(\frac{\text{Cash Inflow}_t}{(1+r)^t} \right)}{\text{Initial Investment}}$$

- **Decision Rule**:
  - If PI > 1, the project is acceptable.
  - If PI < 1, the project should be rejected.
- **Strengths**: Useful for comparing projects of different sizes, especially when capital is constrained.

5. Accounting Rate of Return (ARR)

- **Definition**: ARR measures the return on investment based on accounting profits, not cash flows. It compares the average annual accounting profit of a project to its initial investment.
- **Formula:**

$$ARR = \frac{\text{Average Annual Profit}}{\text{Initial Investment}}$$

- **Decision Rule**:
  - Accept the project if ARR exceeds the company's required rate of return.
- **Strengths**: Simple to compute and based on accounting figures, which are easily accessible.
- **Limitations**: Ignores the time value of money and focuses only on accounting profit, not cash flow

Each capital budgeting technique has its strengths and weaknesses, and businesses often use a combination of these methods to evaluate investments. **NPV** and **IRR** are generally the most reliable techniques, as they take into account the time value of money and provide a clear indication of a project's potential profitability. However, techniques like **payback period** and **ARR** can still offer valuable insights, especially in simpler scenarios or when a business is constrained by liquidity or resources.

Problems:

1. Net Present Value (NPV)

Problem:

You are considering an investment project that requires an initial investment of $1,000. The project is expected to generate $300 in cash inflows at the end of each year for 4 years. The required rate of return is 8%. Should you accept the project?

Solution Steps:

- Initial investment: $1,000
- Cash inflows: $300 per year for 4 years
- Discount rate (r): 8%

Use the NPV formula:

$$NPV = \sum \left(\frac{CF_t}{(1+r)^t} \right) - InitialInvestment$$

Where CF_t is the cash inflow at time t.

$$NPV = \frac{300}{(1+0.08)^1} + \frac{300}{(1+0.08)^2} + \frac{300}{(1+0.08)^3} + \frac{300}{(1+0.08)^4} - 1000$$

Calculate the NPV and decide if it's positive (accept the project) or negative (reject the project).

2. Internal Rate of Return (IRR)

Problem:

You are evaluating a project with an initial investment of $2,000. The project generates $800 in cash inflows each year for 3 years. What is the IRR?

Solution Steps:

- Initial investment: $2,000
- Cash inflows: $800 per year for 3 years

IRR is the rate r where the NPV = 0. So, you solve this equation:

$$0 = \frac{800}{(1+r)^1} + \frac{800}{(1+r)^2} + \frac{800}{(1+r)^3} - 2000$$

This requires either trial and error or the use of a financial calculator or spreadsheet software to find the value of r. The IRR is the rate that makes the NPV = 0.

3. **Payback Period**

Problem:

A company is considering a project that requires an investment of $1,500. The project will generate $500 in cash inflows each year. How long will it take to recover the initial investment?

Solution Steps:

- Initial investment: $1,500
- Annual cash inflows: $500

The payback period is the time it takes to recover the initial investment.

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflows}} = \frac{1500}{500} = 3 \text{ years}$$

So, the payback period is 3 years.

4. **Profitability Index (PI)**

Problem:

An investment project requires an initial investment of $1,200. The expected cash inflows are $400 per year for 5 years. The discount rate is 10%. What is the Profitability Index (PI)?

Solution Steps:

- Initial investment: $1,200
- Cash inflows: $400 per year for 5 years
- Discount rate: 10%

The Profitability Index (PI) is calculated using the formula:

$$PI = \frac{\text{Present Value of Cash Inflows}}{\text{Initial Investment}}$$

First, calculate the present value of cash inflows:

$$PV = \frac{400}{(1+0.10)^1} + \frac{400}{(1+0.10)^2} + \frac{400}{(1+0.10)^3} + \frac{400}{(1+0.10)^4} + \frac{400}{(1+0.10)^5}$$

Then, calculate the PI:

$$PI = \frac{PV}{1200}$$

If PI > 1, the project is considered profitable.

5.Accounting Rate of Return(ARR)

You are evaluating a project that requires an initial investment of $10,000. The project is expected to generate $2,000 in average annual accounting profits (net income) over the next 5 years.

What is the Accounting Rate of Return (ARR) for this project?

Solution Steps:

1. Formula for ARR:

$$ARR = \frac{\text{Average Annual Accounting Profit}}{\text{Initial Investment}} \times 100$$

2. Given:
   - Initial Investment = $10,000
   - Average Annual Accounting Profit = $2,000
3. Calculation:

$$ARR = \frac{2000}{10000} \times 100 = 0.20 \times 100 = 20\%$$

Answer:

The ARR for this project is 20%.

Problems for Self-Learning:

1.A company is considering two investment opportunities A and B that cost Rs.4,00,000 and Rs.3,00,000 respectively. The first project generates Rs.1,00,000 a year for 4 years. The second generates Rs.6,00,000, Rs.1,00,000, Rs.80,000, Rs.90,000 and Rs.70,000 over a five years period. The company's cost of capital is 8%. Which project would you choose under NPV method .

2.Coastal software Ltd.is proposing to mechanize their operations. Two proposals M and N in form of quotations have been received from two different vendors. The proposal in each case costs Rs.5, 00,000. A discount factor of 12% is used to compare the proposals. CFAT are likely to be as under:

Year	Proposal M	Proposal N
1	1,50,000	50,000
2	2,00,000	1,50,000
3	2,50,000	2,00,000
4	1,50,000	3,00,000
5	1,00,000	2,00,000

Which one do you recommend under:
a) Payback period method.
b) Accounting Rate of Return method.
c) Net present value method.

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